

COGNITIVE REHEARSAL IN THE TREATMENT OF OBESITY: A COMPARISON AGAINST CUE AVOIDANCE AND SOCIAL PRESSURE

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Abstract—The effects of a cognitive (cognitive rehearsal) in the treatment of obesity were evaluated in a 16-week weight reduction programme by comparison against two other treatments, cue avoidance, and social pressure. Fifty-three obese females, categorised (on the basis of their weight loss after 3 weeks of dieting) as either high losers or low losers, were randomly assigned to one of these three treatments. Posttreatment weight losses were significantly greater for high losers than low losers, and for subjects in cue avoidance than those in social pressure. Those in cue avoidance went over their diet less often than others, and reported greater changes in eating habits, and in the cognitive domain. Neither subject type nor treatment type has a significant effect on weight change during the follow-up year. High losers adhered to the diet more initially, but also, throughout treatment, lost more weight for the same degree of dietary adherence. Behavioural and physiological characteristics underlying this important dimension of individual differences were discussed.

The evidence of the involvement of cognitive factors in the control of eating, even in treatments which do not specifically attempt to modify them systematically (Bennett, in press-a; Drownowski, 1983), suggests the value of making these the explicit target of treatment. Three studies which have attempted to produce cognitive changes through cognitive techniques (Block, 1980; Collins, 1980; Dunkel & Glaros, 1978) found different results, which may reflect their different lengths of treatment and drop out rates.

In a dismantling study, Bennett (in press-b) compared individualised cognitive rehearsal against three control treatments and found that it produced greater losses, but the superiority failed to reach statistical significance. However the results raised the possibility of a treatment effect for subjects who had lost least weight during the initial 3 weeks of treatment, when they had only received dietary advice.

The main purpose of the present study was to evaluate this suggestion of a subject-treatment interaction by examining the responses to cognitive rehearsal of subjects with either low or high initial weight loss. In doing so, it attempted to clarify the predictive value of initial weight loss. Several other studies have reported that, unlike pretreatment subject characteristics, early response to treatment was a good predictor of outcome (Jeffery, Wing, & Stunkard 1978; Wing & Epstein 1981; Wing, Epstein, & Shapira 1982). The possibility that initial loss may be a subject characteristic that could interact with treatment characteristics was also raised by the results of Brightwell and Naylor (1979) and Craighead (1984).

One interpretation of the predictive value of initial weight loss is that it reflects a stable biological characteristic such as metabolic rate (e.g., Garrow, Durrant, Mann, Stalley, & Warwick, 1978). This "biological hypothesis" suggests that initial weight loss predicts subsequent losses because it reflects a biological tendency to loss weight quickly.

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A second possibility is that the motivating effects of initial weight loss reinforce dieting efforts, so that initial success leads to continued dietary adherence (and initial failure leads to declining adherence). This "motivational hypothesis" seems to have been the rationale of attempts to influence initial weight loss through attempting to boost initial dietary adherence (Wing & Epstein, 1981; Wing, Epstein, & Shapira, 1982). A third hypothesis, the "ability hypothesis," suggests that initial loss predicts subsequent losses because it reflects dieting ability. Dietary adherence is multiply determined and not easily predicted, but a sample of it permits some prediction: behaviour predicts behaviour. Measures of dietary adherence are included in this study in order to evaluate these three hypotheses and to examine this dimension of individual differences.

The study also provided an opportunity to assess the predictive value of externality, as measured by Rodin (e.g., Rodin, Slochower, & Fleming 1977), and to assess wider emotional changes in treatment. In order to assess the effects of cognitive rehearsal, cue avoidance was used as a comparison treatment. This, based on the "food management" programme of McReynolds, Lutz, Paulsen, and Kohrs (1976), is a well-established programme, based on a clear rationale, and of proven efficacy.

METHOD

Subjects

Subjects were selected from referrals by medical practitioners to a weight reduction programme if they fulfilled the following criteria. They had to be: (a) female; (b) aged between 18 years and 60 years; (c) at least 6.8kg overweight according to the Fogarty criteria (Bray, 1975); (d) not currently receiving any treatment for obesity; (e) not diagnosed as suffering from any disease or disorder which might affect weight; (f) not taking any form of medication that would affect water retention, appetite, or metabolism; (g) not pregnant.

The 48 subjects completing treatment had a mean age of 40.0 years, a mean weight of 81.7kg, a mean percentage overweight of 48% (using Fogarty Conference criteria, Bray, 1975) and a mean Body Mass Index (BMI) of 32.6 kg/m².

Measures

Major assessments were carried out at the pretreatment assessment session (PT), at week 16 (the end of treatment) (W16), and at the year follow-up (F3). Other measures were taken continuously during treatment.

Weight. Subjects were weighed privately, in indoor clothing, to the nearest 0.1kg. on a beam balance scale (Marsden Balance Type 150).

Assessment of diet-related characteristics. Eight measures of factors related to dieting were administered at several points during treatment and follow-up.¹ The *Eating Survey Schedule (ESS)* is a measure of perceived difficulty of dieting, or self-efficacy expectations. This requires subjects to rate (on a seven-point scale) how difficult they would find it to control their eating in each of 27 situations. The total is taken. [Administered at PT, week 11 (W11), W16, F3]. The *Eating Thoughts measure (ET)* assesses frequently occurring thoughts about dieting, and is modelled on the self-state-ment tests used by, for example, Schwartz and Gottman (1978). It consists of 15 negative, or maladaptive, and 15 positive, or adaptive, thoughts about dieting. Sub-

¹Copies of the unpublished assessment instruments and treatment manuals devised for the study are available from the author.

jects rate the frequency of occurrence of each type of thought during the previous week using a five-point scale. Separate positive and negative totals are computed [week 4 (W4), W11, week 15 (W15)]. The *Eating Behaviour measure (EB)* is an 18-term self-report of eating patterns over the previous week (PT, W4, W11, W15). The *Target Rating Sheet* was concerned with each subject's target situation—that in which they had experienced most difficulty in controlling their eating between week 1 and week 4. Using an 11-point scale, subjects estimated the probability that they would control their eating in the target situation, if faced with it during the following week (W4, W16). The *Motivation Questionnaire (MQ)* is a 12-item measure of self-reported motivation to lose weight (PT, W16, F3). The *Diet Diary* is a daily record which subjects filled in throughout treatment, categorising each day's dieting efforts as having: (a) adhered to; (b) ate more; or (c) less, than the allowance. The *Restraint Questionnaire (RQ)* of Polivy, Herman, and Warsh, 1978 was employed (PT, W16, F3). The *Treatment Outcome Expectations Sheet (TOES)* assesses subjects' expectations of (a) their total weight loss (in pounds) at week 16, (b) their amount of self-control at week 16, and (c) the amount of benefit to be derived from the programme (W1, W4, W8, W11).

Assessment of wider psychological effects of treatment. The following questionnaires were administered at PT, W16, and F3: The *Middlesex Hospital Questionnaire (MHQ)* (Crown & Crisp, 1979); The *Eysenck Personality Questionnaire (EPQ)* (Eysenck & Eysenck, 1975); and *Locus of control orientation (IE)* which was assessed using a 5-item questionnaire embodying the Personal Control Factor obtained by Gurin, Gurin, Lac, and Beattie (1969).

Subjects reports of change and evaluation of treatment. On W16, subjects were asked to estimate how much they had changed since the beginning of treatment in each of 18 specific areas, of which 6 were targets of Cue Avoidance (CA) (e.g., "changed way of storing foods"), 6 targets of Cognitive Rehearsal (CR) (e.g., "Increased ability to say 'no' to self") and 6 common to all three treatments (e.g., "increased knowledge of food"). These ratings were made on seven-point scales.

Assessment of life change. This was carried out through a structured interview using a list of 82 closely defined life-events (Callaghan, 1976). Consensual ratings of the amount of change represented by each event had been made by 100 women attending commercial slimming clubs, and means of these ratings were used as life change unit (LCU) scores. Life change was assessed on W16 regarding the treatment period and at F3 regarding the follow-up year.

Possible predictors. Five possible predictors were assessed at PT. Three measures of externality comparable to, and based on, those employed in previous research by Rodin (e.g., Rodin et al., 1977) were a *time estimation task (TET)*, a *slide rating task (SRT)*, and an *immediate recall task (IRT)*. The synonym subtests of the two Senior Forms of the *Mill-Hill Vocabulary Scale (MHV)* (Raven, 1958) were employed as a measure of verbal intellectual functioning. The *Food Value Questionnaire (FVQ)* is a 20-item forced choice measure of knowledge of calorific values.

Procedure

Potential subjects attended for assessment on PT and those eligible were invited to attend a large initial meeting the following week (W1), at which they were given dietary advice to follow on their own for 3 weeks until the second large meeting on W4. At this second meeting subjects were classified on the basis of a median split of their weight

loss in the interval and randomly assigned from within each of these two categories to one of three treatment programmes, (a) CR, (b) CA, or (c) Social Pressure (SP). Treatment groups met on weeks 5, 6, 7, 8, 9, 11 and 15 and all subjects had an individual session on W16. After posttreatment assessment at W16 attempts were made to follow-up all subjects 3, 6 and 12 months later (F1, F2, F3).

All sessions were conducted by a qualified male clinical psychologist with 3 years' experience of behavioural treatments of obesity. An experienced dietitian played a major part in the session on W4 and acted as nutritional consultant to the programme. All subjects received nine 75-minute long, group sessions, and filled in Diet Diary forms daily.

The programme common to all subjects: Week 1, week 4, week 16 and follow-up. On W1 subjects were given a diet follow on their own until W4. An Anglicised version of the exchange system described by Stuart and Davis (1972), and developed by Warrington (1977), was introduced. Subjects were given specific "quotas" of exchanges, effectively representing calorie levels 1000Kcal below expected energy requirements, with a minimum of 1000Kcals. Diary forms and stamped addressed envelopes were provided. On W4 the opening review of weight losses was followed by a talk on obesity and dieting given by the dietitian. Then the arrangements for the small treatment groups were announced (the subjects having been categorised and randomly allocated by the psychologist during the talk). Subjects identified and rated their target situation. The session on W16 was largely used for posttreatment assessment. Followup sessions were arranged individually: nonattendance was followed (where appropriate) by a letter, a telephone call, and a visit to the subject's home.

CA. This consisted of specific detailed advice recommending subjects to reduce their exposure to food cues by making a variety of changes in their habits, and was based on the food management programme of McReynolds *et al.*, 1976. On W5, subjects were introduced to the rationale of "making it easy for yourself" and reducing "temptation" through reducing contact with food cues. There was detailed discussion of changes to make in food storage habits, summarised in a handout. Next came a discussion of a common target problem (coming home from work, feeling tired and immediately eating readily available food), as practice in approaching problems from the point of view of CA. Subsequent sessions followed the same format of (a) a brief review of recent dieting efforts, (b) a central 'lesson' giving specific detailed advice on ways of reducing contact with food and one area of food management, (c) a discussion of a specific problem from the point of view of the programme as practice in problem solving, and (d) a summary of the content of the session. On W15 the advice was reviewed and summarised.

CR. On W5, subjects were presented with the rationale that control of overeating could be increased through learning to resist, rather than avoid, temptation: This was to be achieved by learning to talk to themselves in more effective ways. CR was introduced as a method of learning how to do this. The format of this (and of the other six sessions) included (a) a brief review of recent efforts, (b) a long period of imaginal rehearsal, and (c) a summary. During the initial imaginal rehearsal, subjects imagined 10 complete scenes as they were described by the psychologist. Each of the scenes (which were based on problems frequently reported in this clinic) consisted of (a) description of the circumstances, (b) description of the feeling of temptation accompanied by a maladaptive cognition, (c) a positive restraining self-statement, (d) subsequent adaptive behaviour and (e) subsequent reinforcement or self-reinforcement.

After an initial presentation, each scenario was then presented again but without (c), (d), or (e), and subjects produced a self-statement and course of action which 'fitted' better for them. The psychologist related each scene through (b) and subjects completed it themselves. After a period of rehearsal the rationale was restated and summarised; homework rehearsal was then allocated and a written handout provided. In subsequent sessions 15 standard scenes were rehearsed, first with subjects being presented with complete versions and then, following some practice, after being presented with a truncated version to which they supplied their own self-statements, responses and self-reinforcements. Subjects were encouraged to produce their own elaborated endings but could use the standard endings if needed. On W15, the use of the technique was reviewed.

SP. On W5, subjects were introduced to the rationale and routine of the treatment. They were told that they possessed basic self-control ability and that the course of meetings would help them to strengthen it. As it became stronger they would become able to do without the group. The meetings would be used to review problems encountered to discuss how to deal with them. The session moved to a discussion of adherence to the diet, and subjects each described problems they would be likely to encounter during the following week. Subsequent sessions followed the same format, beginning with a review in which each member reported on her dietary adherence since the previous session. The psychologist encouraged members to comment on this, to praise improvements, and to discuss problems that had occurred or were imminent. He chaired discussion, and praised dietary adherence, but did not evaluate subjects' suggestions, or give advice beyond the assertion that 'the more you stick to the diet, the easier it will become'. On W15, progress was reviewed.

RESULTS

Attrition

Of the 62 eligible subjects, 53 attended on both W1 and W4, 48 on W16 and 40 at F3. The use of Fisher's Exact Probability Test found no evidence of significant differential attrition.

Preliminary analysis

Oneway analyses of variance (ANOVAs) were carried out across treatment conditions on each of a range of pretreatment characteristics, but found no significant difference on any.

Posttreatment results

Unless otherwise stated, the results were analysed using repeated measures ANOVAs or, for change scores, two-way ANOVAs, with subject type and treatment type as factors, and computing the unweighted means solution. Scheffe tests were used for post-hoc testing.

Obesity. Changes in aspects of obesity were assessed in terms of reductions in weight, percentage overweight and Body Mass Index (BMI), and also in terms of the Weight Reduction Index (WRI) and Feinstein's Weight Reduction Index (FI) (Wilson, 1978). These are summarised in Table 1 and depicted, for WRI, in Figure 1. The analysis of the effects of the experiment on weight² over three occasions (W4, W11, and W16) found a main occasions effect ($F = 40.52$, $df = 2,84$, $P < 0.001$). There were

²Parallel analyses were carried out on percentage over weight and BMI, but since the results were identical to those of weight, only the latter are reported.

Table 1. Means (and standard deviations) of indices of obesity reduction during treatment and follow-up.

Subject Grouping	Occasion	n	Indices of Obesity Reduction				
			Weight Loss	% Loss	BMI Loss	WRI	FI
CA-LL	W4	8	1.8(0.8)	3.2(1.6)	0.7(0.3)	6.0(15.7)	9.2(16.6)
	W16	8	5.8(3.9)	10.6(7.1)	2.3(1.5)	29.0(27.5)	39.6(32.5)
	F3	6	3.2(10.9)	5.9(18.3)	1.2(4.0)	33.1(59.4)	39.1(72.0)
CA-HL	W4	7	3.7(0.6)	6.5(0.9)	1.4(0.2)	12.7(6.0)	19.2(6.2)
	W16	7	9.6(2.0)	16.9(3.4)	3.6(0.7)	32.8(14.9)	49.8(16.1)
	F3	6	5.3(5.5)	9.2(10.1)	2.0(2.2)	22.2(24.1)	31.4(33.6)
CR-LL	W4	8	1.2(1.7)	2.2(3.1)	0.5(0.7)	7.0(8.1)	9.2(10.6)
	W16	8	1.2(4.2)	2.2(7.8)	0.5(1.7)	8.3(11.2)	10.5(18.5)
	F3	7	0.2(3.4)	0.5(6.5)	0.1(1.4)	3.3(15.6)	3.7(21.2)
CR-HL	W4	8	3.6(1.1)	7.1(2.2)	1.6(0.5)	15.4(12.5)	22.5(12.8)
	W16	8	8.5(3.5)	16.5(6.9)	3.7(1.5)	40.8(44.9)	57.3(49.1)
	F3	7	5.4(3.7)	10.6(7.4)	2.4(1.7)	28.5(41.0)	39.2(46.1)
SP-LL	W4	8	1.1(1.0)	2.1(1.8)	0.4(0.4)	8.6(9.9)	10.7(11.3)
	W16	8	2.2(2.4)	3.9(4.3)	0.8(0.9)	14.5(12.2)	18.4(16.2)
	F3	6	0.4(3.6)	0.7(6.3)	0.1(1.4)	5.5(17.0)	6.2(23.1)
SP-HL	W4	9	4.0(1.5)	7.0(2.2)	1.5(0.5)	14.9(4.3)	21.9(4.9)
	W16	9	5.8(4.2)	10.6(1.6)	2.3(1.6)	25.1(15.4)	35.6(22.2)
	F3	8	4.3(5.2)	7.3(8.5)	1.6(1.9)	13.6(14.6)	20.8(22.2)
ALL	W4	48	2.6(1.7)	4.7(3.0)	1.0(0.6)	10.8(10.3)	15.5(12.1)
	W16	48	5.4(4.5)	9.9(8.3)	2.2(1.8)	24.9(25.4)	34.9(31.5)
	F3	40	3.2(5.8)	5.8(10.2)	1.2(2.2)	17.4(32.0)	23.2(39.6)

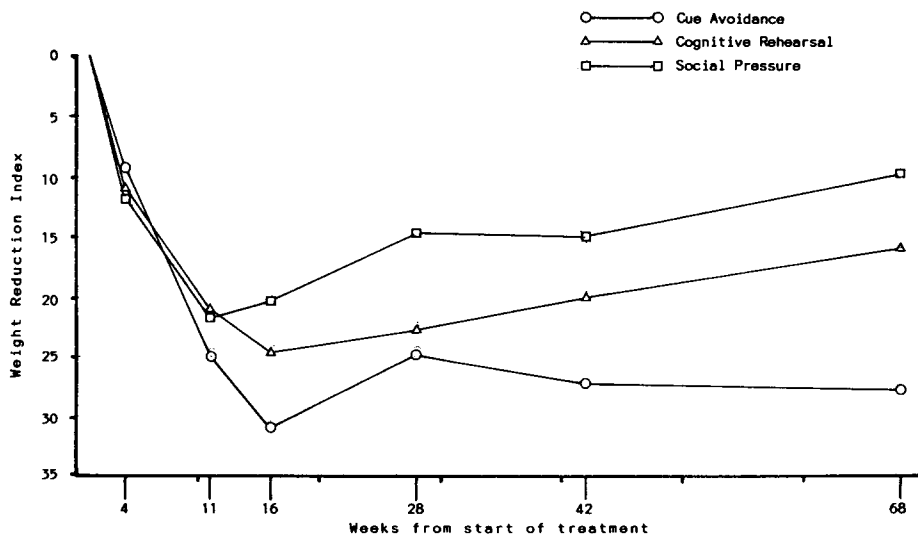


Fig. 1. Weight reduction index for treatment conditions across 16 weeks of treatment and 52 weeks of follow-up.

significant differences between all pairs of occasions, bar W11 and W16. There were significant two-way interactions between occasions and both subject types ($F = 6.69$, $df = 2,84$, $P < 0.01$) and treatment types ($F = 5.06$, $df = 4,84$, $P < 0.01$) but no significant three-way interactions. High losers (HLs) lost significantly more than Low losers (LLs), and subjects in CA lost significantly more than those in SP. In the analyses of WRI and FI between W4 and W16, the only significant effect was for the subjects factor for FI ($F = 4.28$, $df = 2,42$, $P < 0.05$) indicating greater losses for HLs.

Diet-related characteristics: EB. Since responses to most items of the EB were extremely skewed, non parametric analyses were employed. The Wilcoxon Matched Pairs Signed Ranks Test was employed to compare scores of items of the EB obtained at W4 and W15. Analyses were conducted within each subject type and treatment type. Significant changes were only found within CA, representing reductions in frequencies in clearing plates during meals ($Z = 2.52$, $df = 13$, $P < 0.05$) and eating when not physically hungry ($Z = 2.37$, $df = 13$, $P < 0.05$).

ET. The analyses across three occasions found no significant effects for negative thoughts, and only one significant effect for positive thoughts, for the occasions factor ($F = 5.04$, $df = 2$, $P < 0.01$), indicating a decline. There were significant differences only between scores for W4 and W15, and between W11 and W15.

Diary. The data for the completed diaries of 47 subjects were reduced by dividing the 15 weeks into 5 periods of 3 weeks, labelled consecutively A, B, C, D, and E. The frequencies of the three outcomes within each period were computed for each subject. The analyses of these three indices across periods B to E inclusive, only found significant effects on the days over the diet. These were on the treatment factor ($F = 3.81$, $df = 2,41$, $P < 0.05$), the occasions factor ($F = 3.50$, $df = 3,123$, $P < 0.05$), and their interaction ($F = 2.67$, $df = 6,123$, $P < 0.05$). There was a general increase in days over the diet with succeeding periods, with significant differences between period E and periods B and C. Subjects in CA went over their diet less than those in CR and SP—who did not differ one from another. Post-hoc examination of the interaction effect (depicted in Figure 2) found no effect across periods for CA but significant effects for CR and SP. Subjects in CR and SP, but not those in CA, increased the number of days that they went over their diet.

Target. Ratings increased significantly from W4 to W16 ($F = 23.86$, $df = 1,37$, $P < 0.001$). The interaction between occasions and subject types was significant ($F = 4.95$, $df = 1,37$, $P < 0.05$). There were changes over occasions for HLs alone, indicating increased ratings. There was also a significant effect for the interaction between occasions and treatments ($F = 3.61$, $df = 2,37$, $P < 0.05$). There were significant changes for CA and CR, but not SP, marking increases in the first two treatments.

ESS. Scores showed a reduction over PT, W11, and W16 ($F = 29.39$, $df = 2,74$, $P < 0.001$) with significant differences between the mean scores for occasions. There was a significant interaction effect between treatments and occasions ($F = 4.12$, $df = 4,74$, $P < 0.001$) with significant reductions across occasions for CA and CR, but not SP.

TOES. The analysis across W4, W8, and W11 found a significant occasions effect for the weight loss expected for W16 ($F = 5.15$, $df = 2,84$, $P < 0.05$): post-hoc testing found significant differences between scores on each occasions reflecting a reduction in

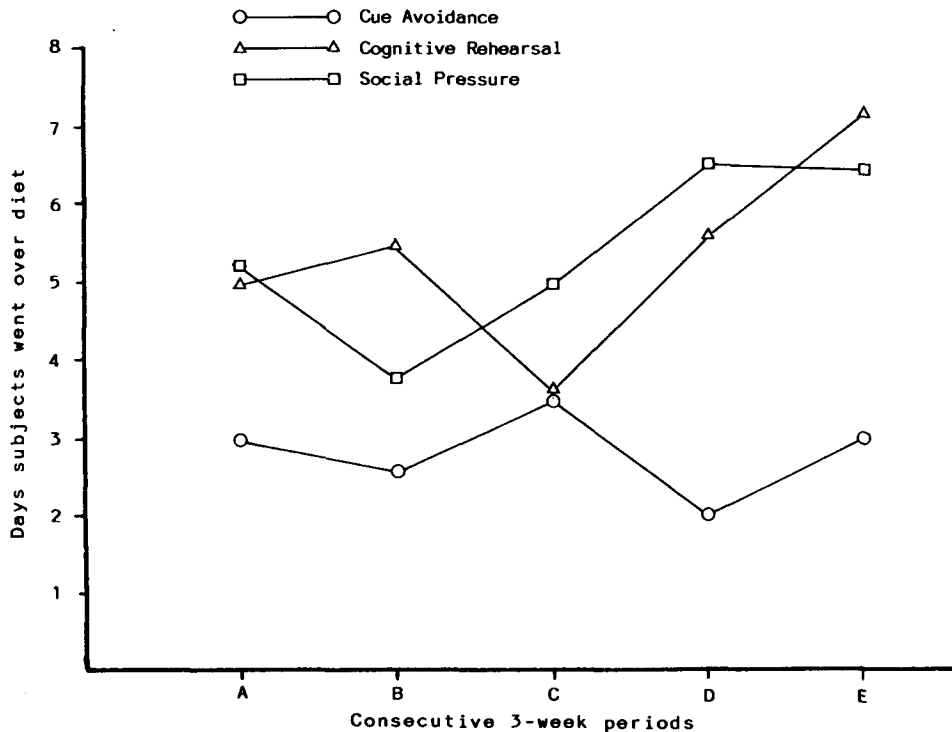


Fig. 2. Numbers of days (within each 21-day period) on which subjects exceeded their dietary allowance.

the overall expected weight loss from 30.71lbs. at W4 to 22.91lbs at W11. There were no other significant effects.

MQ, RQ. The analyses found no significant effects.

Wider psychological changes. The analyses of the EPQ, MHQ, and IE across PT and W16 only found significant occasions effects on 2 of these 11 measures, indicating decreases in MHQ-H ($F = 5.47$, $df = 1,37$, $P < 0.05$) and EPQ-N ($F = 19.36$, $df = 1,37$, $P < 0.001$).

Subjects' Ratings of Change. The ratings of specific changes during treatment were subjected to a PA2 principal factor analysis followed by a VARIMAX rotation (Nie, Hull, Jenkins, Steinbrenner, & Bent, 1975). This yielded four orthogonal factors with eigenvalues exceeding one, all of which were meaningful and which are summarised in Table 2. Because of their loadings, these are labelled consecutively "cognitive restraint of eating," "insight and thinking," "mealtime behaviour change," and "nonmealtime behaviour change." Factor scores were subjected to two-way ANOVAs. The analysis of Factor 1 found significant effects for treatments ($F = 5.16$, $df = 2,37$, $P < 0.05$) and subjects ($F = 10.00$, $df = 1,37$, $P < 0.001$), and a significant interaction effect ($F = 4.82$, $df = 2,37$, $P < 0.05$). Higher change scores were reported by HLs, and there were significantly larger changes in CA than SP. Further analysis of the interaction effect found a difference between subject types only in SP, with HLs changing more. There were no significant effects for Factor 2 or Factor 4 but there was a significant

Table 2. Results of factor analysis of changes reported at week 16.

Variables	Factor Loadings*			
	F1	F2	F3	F4
Increased "inner voice" telling me not to eat	0.88	—	—	—
Increased ability to resist temptation	0.81	—	—	—
Increased ability to say "no" to myself	0.78	—	—	—
Increased self-control	0.75	0.36	—	—
Changed way of thinking when tempted	0.60	—	—	—
Cut down on sweet foods	0.35	—	—	—
Changed how I think about dieting	—	0.81	—	—
Increased understanding of what makes me eat	—	0.75	—	—
Changed the way I think when over-eating	—	0.74	—	—
Changed the ways of preparing and cooking food	—	0.65	—	—
Plan meals more	—	0.43	—	0.74
Increased concentration on flavour of food	—	—	0.65	—
Slowed down eating speed	—	—	0.62	—
Changed way of clearing away food after meals	—	—	0.56	0.47
Changed way of storing food	—	—	0.51	—
Changed shopping habits	—	—	—	0.68
Eigenvalue	5.89	1.67	1.61	1.08
Percentage of variance accounted for	54.2	15.4	14.8	9.9

*Only factor loadings significant at the $P < 0.01$ level are listed (Burt and Banks, 1947).

treatments effect for Factor 3 ($F = 7.55$, $df = 2,37$, $P < 0.01$): CA had larger scores than SP.

Follow-up results

Obesity. In the analysis of the effects of weight over four points of follow-up (W16, F1, F2, F3) the only significant effect was for occasions ($F = 6.34$, $df = 3,96$, $P < 0.001$). There were significant differences between all pairs of occasions bar F1 and F2. There were no significant effects on WRI or FI over follow-up.

Psychological variables. The only significant changes from W16 to F3 among MHQ, EPQ, and IE were the occasions factors for EPQ-N ($F = 4.72$, $df = 1,23$, $P < 0.05$), and MQ ($F = 5.23$, $df = 1,23$, $P < 0.05$), representing a general increase in EPQ-N and a general decrease in MQ scores over follow-up.

PRETREATMENT CHARACTERISTICS AND OBESITY REDUCTION

Correlations were computed between pretreatment characteristics (including potential predictors) and indices of obesity reduction, but only some 10% of correlations reached significance, and the patterns of correlations were random.

Processes of change and outcome

Correlations were computed between the four factored change scores and weight loss at W16: only that for Factor 1 was significant ($R = 0.52$, $df = 43$, $P < 0.001$). A similar set of analyses carried out with measures of obesity reduction over the follow-up year found only Factor 2 to be related to obesity reduction $R = 0.45$, $df = 39$, $P < 0.01$). Obesity reduction at Week 16 was consistently, but weakly, negatively related to reduc-

tion over the follow-up: the correlation between the two was -0.37 ($df = 40$, $P < 0.05$).

Dietary adherence and weight loss

The "ability" hypothesis predicts that (a) dietary adherence is correlated with weight loss, (b) dietary adherence in A is correlated with subsequent adherence, and (c) that during A HLs adhere to their diet more closely than LLs. Prediction: (a) was confirmed by the finding that weight loss was significantly correlated with days over the diet, both in A ($R = -0.36$, $df = 47$, $P < 0.01$) and in B,C,D,E ($R = 0.51$, $df = 47$, $P < 0.001$) as was true for days under the diet, both in A ($R = 0.35$, $df = 47$, $P < 0.01$), and in B,C,D,E ($R = 0.35$, $df = 47$, $P < 0.01$); (b) was confirmed by the finding that, over all subjects, there were significant correlations between indices of adherence in A and those in B,C,D,E, and that the values were, for days over, 0.59, for days under, 0.54, and for days kept to the diet, 0.60 (in each case $df = 47$, $P < 0.001$); and (c) was tested by a oneway ANOVA of indices of dietary adherence in A across subject types. HLs went over their diets less often ($F = 5.30$, $df = 1,46$, $P < 0.05$), but there were no other significant differences.

The "biological" hypothesis predicts that (d) if weight loss at W16 is regressed on indices of dietary adherence within each subject type, there will be differences in intercept, but not in slope, and (e) that HLs will be younger and more obese. Prediction (d) was tested by computing regression equations for each subject type for weight loss at W16 on days gone over the diet from A through E. The equation was significant for both LLs ($F = 5.28$, $df = 1,22$, $P < 0.05$) and HLs ($F = 17.20$, $df = 1,21$, $P < 0.001$). The slope for LLs was -0.10 (standard error 0.04) and that for HLs was -0.17 (standard error 0.04): the intercept for LLs was 5.75 (standard error 1.39) and that for HLs was 11.20 (standard error 0.96). The significance of differences between HLs and LLs in these indices was tested by t -tests (following Davies, 1958). The values of t for slopes and intercepts were, respectively, 0.92 (ns) and 5.92 ($df = 47$, $P < 0.001$), confirming prediction (d). Prediction (e) was tested by comparing pretreatment obesity and age of HLs and LLs. There were no differences in age, but HLs were more obese, there being significant differences in weight ($F = 4.93$), excess weight ($F = 6.71$), percentage overweight ($F = 6.46$), and BMI ($F = 5.08$) (in each case $df = 1,46$, $P < 0.05$).

The "motivational" hypothesis predicts (f) a significant correlation between weight loss at W4 and increases in adherence between A and B. This was tested for each index of adherence by computing the correlation between the difference in adherence in A and B and weight loss at W4. The values were, for days over the diet, -0.32 ($P < 0.01$), for days under the diet, 0.02 (ns), and for days kept to the diet 0.25 ($P < 0.05$) (in each case $df = 47$). These results support prediction (f).

DISCUSSION

The main purpose of the study was to test the suggestion that CR would produce greater weight losses for LLs. There was a clear failure to find a subject-treatment interaction: indeed the pattern of results was in the opposite direction, with HLs losing more in each condition. The reasons for the discrepancy between these findings and those of Bennett (in press-b) are unclear, since similar subjects were treated in the same clinic by similar personnel, over the same period of time, and using very similar procedures. The major difference was that in the former study CR was administered individually, and the nature of the scenes was completely individualised. The content of

the scenes employed was almost identical in each study, and it is hard to account for the discrepancy in terms of treatment mode.

CR was no more effective than the 'inert' SP programme in producing dietary adherence and weight loss. The size of loss from CA (at 7.6 kg) was similar to those of other food management programmes reported in the literature (mean post treatment losses of 8.4 kg by Beneke, Paulsen, McReynolds, Lutz, & Kohrs, 1978, and 7.8 kg by Beneke & Paulsen 1979). CA also produced wider changes—on both eating behaviour and efficacy expectations. Reported behavioural and cognitive changes were greatest in CA. Those in CR did not lose more weight than those in SP, but, unlike them, did increase their efficacy expectations. CR had the limited effect of making subjects feel that dieting had become easier, but did not help them to diet more effectively. Although the "cognitive" treatment failed to produce behavioural change, the "behavioural" treatment produced both behavioural and cognitive change. Although the cognitive procedure was not markedly effective, there was evidence for the involvement of cognitive factors in the study from self-report and correlational data, most importantly the preeminence of "cognitive restraint of eating" among self-reported changes, and the correlation of this factor with outcome.

Contrary to expectations, HLs improved more than LLs in terms of weight reduction, increases in efficacy expectations, self-reported increases in cognitive restraint of eating, and dietary adherence during period A (but not subsequently). This dimension of individual differences appears to have important consequences, and to predict outcome more strongly than most pretreatment characteristics, including the potential predictors included in this study. On theoretical grounds, highly externally responsive dieters might be expected to benefit most from CA, but the results of this study provided no confirmation of this.

The support for each of the three hypotheses about the nature of the predictive character of initial weight loss points to a complex pattern of interrelationships between dietary adherence and weight loss. LLs suffered the double handicap of being less able to control their eating early on, and of losing weight more slowly for the same degree of dietary adherence. Motivational consequences of these two phenomena may be expected to retard efforts in the latter stages of treatment. These results underline the importance of including measures of dietary adherence in studies of behavioural treatments which have their principal effects on behaviours, which are only loosely linked to biological changes, such as weight loss.

These results have clinical implications for selection for treatment, and for the services required by LLs. These dieters might well benefit from explicit education about realistic rates of weight loss, and other efforts to help boost and sustain motivation. Initial weight loss has effects which persist throughout treatment, and maximising initial losses may prove to be a productive tactic. Attempts to implement it have, to date, had rather limited effects, whether they used drugs (e.g., Brightwell & Naylor 1979), initially stringent dietary restriction (e.g., Wing & Epstein 1981) or the initial use of the Scarsdale Diet (Wing, Epstein, & Shapira, 1982). These limited effects may reflect the distance between the methods adopted to produce initial losses and those used to produce either subsequent losses or the maintenance of losses. There is little scope for generalisation effects. It may prove more effective to use methods which are also of value in the later stages of treatment; a possible candidate would be increasing physical activity.

These findings suggest that maximising initial weight loss may be a valuable clinical treatment tactic: maximising initial improvement may also prove to be a generally

valuable treatment approach. The success of, for example, "rapid treatments" for enuresis, stuttering, or nervous habits (Azrin & Nunn, 1974; Azrin, Sneed, & Foxx, 1974) may partially reflect their tactic of immersing patients in a short, concentrated period of therapy. Few studies have examined the predictive value of initial response to treatment, but those which have done so have confirmed its value in such diverse areas as the treatment of agoraphobia (Mathews, Gelder, & Johnston, 1981), stopping smoking (Raw & Russell 1980), and the maintenance of gains in the treatment of enuresis (Finley, Rainwater, & Johnson, 1982).

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